Shuyu (William) Zhang

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EDUCATION

University of Pennsylvania — M.S.E. in Mechanical Engineering *Expected May 2027*

Relevant Coursework: MEAM 6200 Advanced Robotics, MEAM 5200 Intro to Robotics, MEAM 5100 Mechatronics; CIS 580 Machine Perception *(In Progress)*, CIS 520 Machine Learning *(In Progress)*

Southwest Jiaotong University – University of Leeds Joint School — B.E. in Mechanical Engineering *June 2025*

GPA: 3.54 / 4.00
Relevant Coursework: Vibration and Control (94/100), Mechatronics and Measurement Systems (95/100), Solid Mechanics (91/100), Computer Applications in Engineering

TECHNICAL SKILLS

Computer Skills: Experienced in JavaScript and Python; skilled in C/C++ programming, with practical application in STM32 projects; proficient in Matlab; comfortable with Git and Linux
Robotics & Control: ROS, SE(3) control, A*, inverse kinematics, vision manipulation
Design and Mechatronics: SolidWorks, STM32, CubeMX, ESP32, Arduino

PROJECTS

Autonomous Quadrotor Navigation — MEAM 6200 Advanced Robotics *Jan – Mar 2026*

Course project · Python · Hardware: Bitcraze CrazyFlie

– Implemented SE(3) geometric controller (Lee et al.) for stable 3D flight in simulation.

– Built 3D path planner with A* search and greedy line-of-sight pruning; applied minimum-jerk polynomial trajectory generator.

– Achieved collision-free flight through 3 obstacle mazes on real hardware (sim-to-real transfer).

Vision-Guided Robotic Manipulation — MEAM 5200 Intro to Robotics *Nov – Dec 2025*

Course project · Python · ROS · Gazebo · Hardware: Franka Emika Panda

– Built autonomous pick-and-place pipeline (perception → planning → execution) on a 7-DOF Franka Panda arm, handling both static objects and a dynamically rotating platform.

– Implemented inverse kinematics solver and motion planner; developed in Ubuntu / ROS / Gazebo, deployed to real hardware addressing sim-to-real calibration gaps.

Hybrid Autonomous & WiFi-Controlled Competition Robot — MEAM 5100 Mechatronics *Nov – Dec 2025*

Course project · Team of 4 · Led mechanical design; supported embedded + software · C / C++ · ESP32-C3 · SolidWorks · Sensors: ToF, VIVE

– As mechanical design lead, designed an Ackermann-steering chassis with rear differential drive in SolidWorks — **the only team out of 36 to choose Ackermann**; the configuration achieves full mobility with just one servo + one drive motor and offers more intuitive manual control.

– Helped develop the ESP32-C3 control system with custom 12V/5V/3.3V power architecture supporting motors, servos, sensors, and WiFi communication.

– Contributed to the dual-mode software organized as a modular state machine — **autonomous mode** (VIVE-based localization, ToF obstacle sensing, PID wall-following, geometric target tracking, button-hit targeting); **manual mode** (web-based WiFi UI for real-time control). Team completed all autonomous tasks.

Multi-Arm Crane Robot — Team Leader, Crane Competition (SWJTU) *Mar – Jul 2024*

Team Leader · C / C++ · STM32 + CubeMX · PID · 23 kg robot, 8 motors + 4 servos + IR sensors

– Directed the electrical control system — programmed and debugged embedded systems (STM32 + CubeMX, C/C++, PID) for 8 motors (2× RM3508 drive + 6× RM2006 across 3 independent rotating arms), 4 servos (3× 15 kg + 1× 50 kg), and infrared sensors; the 23 kg robot completed the multi-angle grip-and-stack task with a 1 kg payload.

– Provided technical oversight for structural wiring; ensured seamless mechanical-electrical integration and participated in mechanical design and multi-process manufacturing (machining + 3D printing + sheet-metal cutting).

Additional Project: Volunteer Tracker *(live · GitHub) — Solo-architected production full-stack platform paired with Claude Code as implementation agent (Next.js / PostgreSQL / Docker / Cloudflare Tunnel); designed a three-layer self-healing ops loop (OpenClaw + SSH + UptimeRobot).*
Industry Experience: ABB Robotics *— Software Engineering Intern, Shanghai (Jan – Feb 2024) — Introductory study of Kalman filter fundamentals (no hands-on implementation); developed early awareness of engineering safety practices in an industrial setting.*